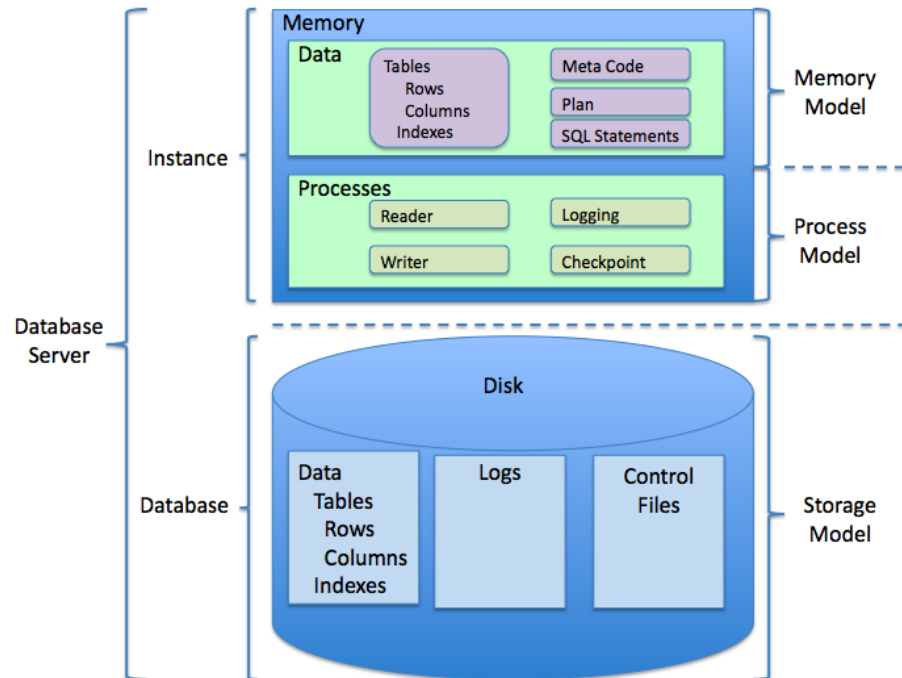


Database Management System



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Steps to follow to Build an ERD

Step 01: Identify Entities (Nouns)

Step 02: Identify Attributes of each Entity

Step 03: Identify relationships between Entities.

Step 04: Draw Global Schema

Problem 01: A company has several departments. Each department has a supervisor and at least one employee. Employees must be assigned to at least one, but possibly more departments. At least one employee is assigned to a project, but an employee may be on vacation and not assigned to any projects. The important data fields are the names of the departments, projects, supervisors and employees, as well as the supervisor and employee number and a unique project number

Problem 02: A salesperson may manage many other salesperson. A salesperson is managed by only one salesperson. A salesperson can be an agent for many customers. A customer is managed by one salesperson. A customer can place many orders. An order can be placed by one customer. An order lists many inventory items. An inventory item may be listed on many orders. An inventory item is assembled from many parts. A part may be assembled into many inventory items. Many employees assemble an inventory item from many parts. A supplier supplies many parts. A part may be supplied by many suppliers.

Problem 03: An organization can have multi-Campus schools. Each campus consists on multiple rooms. Each room can either be used as a lecture theatre or a lab. A school can have multiple departments and each department can run multiple programs. Each program consists on multiple courses. At the start of each session, a teacher who is a part of a department is assigned multiple courses from multiple programs to teach one or many classes. Each class belongs to a particular program and level but each program and level may have multiple classes in it.